Joe Sanchez

Software Engineer

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Bothell, WA

Summary

Skilled Software Engineer with experience developing cloud infrastructure using AWS/Terraform, along with developing applications in Spring, React, .NET, and three.js. Proficient in multiple programming languages, including Java, JavaScript, Python, and C#. Adept at developing backend infrastructure, integrating secure authentication, building simulation systems, and contributing to content management systems. Looking to leverage my skills to contribute to innovative projects and integrate myself into a robust team where I can learn from my peers and continue my journey as a Software Engineer.

Work Experience

Independent Software Development

Personal Projects, Bothell, WA — December 2025 - Present

- Design, build, and deploy full-stack web applications end to end, including City Builder, Chess, Triage, and Tone Radar (see Projects), each live on its own subdomain.
- Provision AWS hosting with Terraform: S3 + CloudFront static sites, Lambda functions behind API Gateway, and Nginx on EC2 behind an Application Load Balancer and Auto Scaling group, with Route 53 DNS and ACM certificates.
- Integrate AI models and third-party services, including TypeSafe Jev for text scoring and job ranking, Apify for job-board scraping, and JWT and Auth0 for authentication.
- Practice data structures and algorithms on LeetCode, with 551 problems solved in total (258 Medium, 22 Hard), mostly in JavaScript and Java.

Software Developer

Tilosi, Bothell, WA — June 2024 - December 2025

- Worked with peers to build core simulation systems for Towns and Wilds using C#, .NET 5, WPF, and a 1024x1024 pixel canvas, providing an immersive and engaging gameplay experience.
- Developed and maintained scalable C#/.NET microservices and RESTful APIs handling persistent game updates and dynamic environment processes.
- Helped develop and implement AI-driven zone systems with behavior trees, procedural generation, and dynamic environment mechanics.

Software Engineer II

Sinclair Broadcast Group, Seattle, WA — August 2023 - June 2024

- Helped design and develop interactive ATSC 3.0 broadcast applications with React, Node.js, and Canvas, creating an immersive and engaging experience to spearhead enterprise broadcast applications.
- Built and maintained backend systems with Node.js and SQL/DynamoDB powered by AWS. Provisioned infrastructure with Terraform and streamlined deployments through CI/CD pipelines.
- Designed and deployed robust AWS Lambda functions that handle user traffic in tandem with API Gateway.
- Collaborated in an Agile team to diagnose and resolve deep-rooted platform issues, enhancing stability and maintainability.

Software Engineer I

Sinclair Broadcast Group, Seattle, WA — March 2023 - August 2023

- Developed a Java Spring data migration service, automating the transfer of millions of SQL records from production to lower environments.
- Debugged and designed features in an Agile team to improve the readability and sustainability of Sinclair's Storyline CMS using Java, Spring, and SQL, while implementing and maintaining OOP best practices.

Freight/Receiving Associate

The Home Depot, Bothell, WA — December 2025 - Present

- Load and unload trucks and move material from the receiving area throughout the store.

Projects

City Builder
city-builder.aws-prac-route53.com — three.js, Vite, Express, MongoDB

- Building a 3D city-building game in three.js with Claude Code (about 25,000 lines of JavaScript) with a road network of curved segments, A* trip routing, and lane-level traffic that queues at traffic lights, give-way junctions, and roundabouts.
- Simulates zoning, building growth, utilities, and a production-chain economy across 13 procedurally generated island maps, with compact saves and an Express/MongoDB account API with JWT auth.

Chess
chess.aws-prac-route53.com — Canvas, Webpack, Lambda, DynamoDB, Terraform

- Built a chess game drawn on HTML5 Canvas with a computer opponent and email challenges, served from S3 + CloudFront on a Route 53 domain.
- Every move is re-validated by a Node.js Lambda behind API Gateway against the board stored in DynamoDB; all infrastructure is provisioned with Terraform.

Triage
triage.aws-prac-route53.com — React, Express, MongoDB, Apify, TypeSafe Jev

- Built a job search that runs LinkedIn, Indeed, and Glassdoor scrapers in parallel, merges duplicate postings, and ranks every posting by fit with an AI model.
- Implemented JWT auth, resume upload and parsing (PDF/DOCX), rate limiting, and Helmet security headers on an Express 5 API.

Tone Radar
tone-radar.aws-prac-route53.com — React, Express, MongoDB, TypeSafe Jev

- Built a PWA that scores every sentence of an email for passive-aggression, blame, hedging, and unclear asks, and suggests rewrites that an AI model checks keep the original meaning.

Skills

Languages: Java, JavaScript, Python, C#, SQL, NoSQL, Bash, PowerShell
Frontend: HTML, CSS, React, Next.js, React Native, Redux, jQuery, Tailwind CSS, Webpack, Vite, UI/UX Design, Accessibility
Backend & Security: Node.js, Express, Spring, Spring Boot, .NET, REST APIs, GraphQL, Microservices, OAuth, JWT, Auth0
Databases: MySQL, SQL Server, MongoDB, DynamoDB
Cloud & DevOps: AWS, Serverless, IAM, Terraform, EC2 (Linux), Linux, ALB, Auto Scaling, S3 + CloudFront, API Gateway, Lambda, Route 53, Nginx, CI/CD, GitLab CI, GitHub Actions, Observability
AI Integration: LLMs, LLM APIs, Generative AI, Prompt Engineering, Machine Learning, AI Tools, TypeSafe Jev, Apify
Graphics & Simulation: Three.js, HTML5 Canvas, WPF, Procedural Generation
Testing: Unit, Integration, Jest, TDD, A/B Testing
Version Control & Process: Git, GitHub, GitLab, Jira, Agile, Scrum
Data & Product: Data Analysis, Data Visualization, Prototyping, Product Management, Project Management, Customer Service

Education

Full-Stack Development Certification
University of Washington, Seattle, WA — 2022 - 2023

Computer Programming, Cloud Architecture
Self-Taught, Woodinville, WA — 2015 - Present

Honors

Letter of Recommendation
Sinclair Broadcast Group, Seattle, WA — Letter of Recommendation
Author: Chris Taylor, Senior Technical Director

Certifications

- AWS SAA-C03 | AWS Certified Solutions Architect - Associate
- AWS CLF-C02 | AWS Certified Cloud Practitioner

Websites, Profiles

Portfolio Website | GitHub Profile | LinkedIn Profile | LeetCode Profile